Exact controls for homometric scattering: proofs and measurement limits

Scientific completion record for the fixed 63-site construction

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Abstract

We complete fresh global mean-crossing certificates for three previously studied Gaussian-motion controls: the original and marginal-preserving controls each have exactly two simple positive crossings, and the harmonic control has exactly one. We then impose a stronger control: both homometric structures receive the same independent Gaussian motion at every site, and their entire centered displacement covariance matrices agree. Their ensemble mean intensities agree at every frequency. At a common clean amplitude null, their first three intensity moments agree exactly, while their fourth moments differ at every finite positive noise level. A general cumulant selection theorem identifies the difference as a known diagonal bispectral invariant times an explicit noise coefficient. For the supplied pair the resulting fourth-moment gap factors into strictly positive terms. Common Gaussian readout noise and Poisson counting preserve identifiability. Exact small-noise Hellinger asymptotics give sharp orders for independent sample requirements: $\Theta(s^{-10})$ for fixed Gaussian readout or fixed positive photon background, and $\Theta(s^{-6})$ for zero-background counting. The proofs concern the specified mathematical models. Publication priority, detector calibration, and physical realization are not established by this record.

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1 What is complete and what the claims mean

The strongest completed statement is that two explicitly specified, noncongruent point sets remain indistinguishable in their mean diffraction under one common displacement law, yet have different intensity distributions at every finite positive noise level. The distinction first appears at the fourth intensity moment at the selected frequency. This is proved for a general class of pairs and verified exactly for the fixed 63-site example.

The earlier crossing problem is now closed for all three original controls. The proof uses exact algebraic coefficients, outward integer intervals, analytic treatment of the origin, finite interval covers, and bounds on the entire infinite tail. It does not infer a global root count from a grid.

The measurement analysis establishes both a possible discriminator and a limitation. Distributional nonidentity survives two specified detector models, but repeated measurements become expensive as motion tends to zero. This conclusion follows from probability distances, rather than the numerical size of an unnormalized moment alone.

Question	Present status
Mathematical correctness	Explicit analytic proofs plus exact-arithmetic certificates and separate internal reviews.
Originality	Established methods and close prior work identified. Priority for the precise theorem and construction remains unverified.
Scientific usefulness	Identifiability, a specific statistic, detector asymptotics, and sharp sample-order limits are proved within stated models.
Physical validation	No measured data, calibrated instrument, or physical implementation is supplied here.

"Proved" in this report means a deductive statement from explicit assumptions, with the computational portions supported by reproducible exact or outward arithmetic. The reviews were performed by separate AI analysis agents. They are useful internal checks, not external peer review or verification in a formal proof assistant. No percentage of infallibility is assigned.

2 The fixed structures and the common-motion model

2.1 Geometry, clean homometry, and the observation frequency

Set

$$\theta = \arcsin\frac{1}{5\sqrt{3}}, \quad \phi = -\arcsin\frac{1}{8\sqrt{3}}, \quad \alpha = \frac{3\theta}{\pi}, \quad \beta = \frac{3\phi}{\pi}.$$

Let U consist of the points $3j + 1 + (-1)^j\alpha$ and $3j + \epsilon_j$ for $j = 0, \dots, 9$, together with 30, where

$$\epsilon_0 = \epsilon_1 = 0, \qquad \epsilon_j = (-1)^j\beta \quad (j = 2, \dots, 9).$$

Define

$$V_A = \{0, 31, 63\}, \quad V_B = \{0, 32, 63\} = 63 - V_A, \qquad A = U + V_A, \quad B = U + V_B.$$

The bounds $0 < \alpha < 1/4$ and $-1/4 < \beta < 0$ show that the listed motif points are distinct. The motif lies in $[0, 30]$, and its three translates are disjoint in each configuration, so both sets have $n = 63$ sites.

For any finite point set G, put $F_G(q) = \sum_{r\in G} e^{iqr}$. Sum-set factorization gives, for every real q,

$$F_A(q) = F_U(q)F_{V_A}(q), \qquad F_B(q) = F_U(q)e^{63iq}\overline{F_{V_A}(q)}.$$

Consequently $|F_A(q)|^2 = |F_B(q)|^2$ for every q: the clean structures are homometric. Choose $q_0 = \pi/3$ and write

$$z_{Gj} = e^{iq_0 r_{Gj}}, \qquad M_{G,k} = \sum_j z_{Gj}^k.$$

Exact phase evaluation gives $M_{A,1} = M_{B,1} = 0$. The required algebra uses

$$e^{i\theta} = \frac{\sqrt{222} + i\sqrt{3}}{15}, \qquad e^{i\phi} = \frac{\sqrt{573} - i\sqrt{3}}{24},$$

and the sixth roots of unity associated with integer base positions. All quantities can therefore be calculated in $\mathbb{Q}(\sqrt{3}, \sqrt{74}, \sqrt{191}, i)$ with rational coefficients.

2.2 Identical displacement laws, including the full covariance

For each configuration, give every site an independent displacement $\xi_j \sim N(0, \tau^2)$, with the same τ^2 for every site of both structures. Optional centering gives

$$u = P\xi, \qquad P = \mathrm{Id} - \mathbf{1}\mathbf{1}^T/n, \qquad \mathrm{Cov}(u) = \tau^2 P.$$

Thus A and B have exactly the same centered covariance matrix. Each marginal variance is $\tau^2(n-1)/n$ and every off-diagonal covariance is $-\tau^2/n$. Centering multiplies the field by a phase of modulus one, so it changes no intensity in any realization.

Let $s = q_0^2\tau^2$ be the raw Gaussian phase variance, and let $v = \mathbb{E}\|u\|^2/n$ be the centered displacement mean square. Then

$$s = \frac{7\pi^2}{62} v, \qquad t = e^{-s/2}, \qquad x = e^{-s} = t^2. \tag{1}$$

The intensity at the selected frequency has the equivalent representation

$$F_G(s) = \sum_{j=1}^n z_{Gj} e^{i\sqrt{s} Z_j}, \qquad I_G(s) = |F_G(s)|^2, \qquad Z_j \overset{\text{iid}}{\sim} N(0,1).$$

In particular, $0 \le I_G \le n^2$.

The distinction between s and a standard deviation matters: all powers in the detector and sampling results below are powers of phase *variance*. The symbol x in (1) is an attenuation in $(0, 1]$. The root certificates later use a different, nonnegative coordinate y.

2.3 The mean remains equal at every frequency

For independent identical displacements with characteristic function $\psi(q) = \mathbb{E}e^{iq\xi_j}$, separating diagonal and off-diagonal terms gives

$$\mathbb{E}I_G(q) = n + |\psi(q)|^2(|F_G(q)|^2 - n). \tag{2}$$

This exact identity, whose displacement-theory background is established [1, 2], does not require Gaussianity. Clean homometry therefore makes the means identical at every frequency. For Gaussian displacements $|\psi(q)|^2 = e^{-q^2\tau^2}$. At q_0 the common mean is $n(1 - e^{-s}) = n(1 - t^2)$.

3 A general theorem selecting the fourth intensity moment

Theorem 1 (Homometric null with a nonzero diagonal invariant)**.** *Let two collections of n unit complex phases satisfy*

$$M_{A,1} = M_{B,1} = 0, \qquad |M_{A,k}| = |M_{B,k}| \quad (k = 2, 3, 4).$$

Apply the same independent Gaussian phase variance s at every site. Define

$$D = \mathrm{Re}\Big(M_{A,4}\overline{M_{A,2}}^{\,2}\Big) - \mathrm{Re}\Big(M_{B,4}\overline{M_{B,2}}^{\,2}\Big).$$

Then the intensity moments of orders zero through three agree for every $s \geq 0$, and

$$\mathbb{E}I_A^4 - \mathbb{E}I_B^4 = 6\kappa_{40}(t)\kappa_{20}(t)^2 D, \tag{3}$$

where

$$\kappa_{20} = t^4 - t^2, \qquad \kappa_{40} = t^4(t^2-1)^3(t^6+3t^4+6t^2+6).$$

If $D \neq 0$, the two intensity distributions differ for every finite $s > 0$.

Proof. For one random phase $w = e^{i\sqrt{s}Z}$,

$$\mu_{ab} = \mathbb{E}(w^a \bar{w}^b) = t^{(a-b)^2}.$$

Let κ_{ab} be its mixed cumulants, defined by the coefficients of $\log \mathbb{E}e^{uw+v\bar{w}}$ near zero. Since $|w| = 1$, the generating function exists there. Deterministic phase multiplication and independence give the field cumulants

$$K_{G,ab} = \kappa_{ab} M_{G,a-b}, \qquad M_{G,0} = n, \quad M_{G,-k} = \overline{M_{G,k}}.$$

The single-site cumulants are real and symmetric in a, b. The ordinary second and fourth cumulant identities give

$$\begin{aligned}
\kappa_{20} &= t^4 - t^2, \\
\kappa_{40} &= t^{16} - 4t^{10} - 3t^8 + 12t^6 - 6t^4 \\
&= t^4(t^2-1)^3(t^6+3t^4+6t^2+6).
\end{aligned}$$

Expand $\mathbb{E}(F_G^p \bar{F}_G^p)$ by partitions of p labelled positive slots and p labelled negative slots. A block containing a positive and b negative slots contributes $\kappa_{ab} M_{G,a-b}$. Call $a - b$ its charge. The charges sum to zero. Any block of charge ± 1 vanishes by the null condition, while a zero-charge block contributes the common structural factor n. The sum of positive charges is at most p.

For $p \leq 3$, every surviving nonzero charge list is a pair $(+2, -2)$ or $(+3, -3)$, possibly accompanied by zero-charge blocks. Its structural dependence is through $n, |M_2|^2, |M_3|^2$, which are shared. For $p = 4$, the complete nonzero-charge possibilities are

$$\begin{gathered}
\varnothing; \quad (+2,-2),\ (+3,-3),\ (+4,-4); \\
(+2,+2,-2,-2); \quad (+4,-2,-2),\ (-4,+2,+2).
\end{gathered}$$

The first five types depend only on the shared magnitudes. In the pattern $(+4, -2, -2)$, all eight slots are used: one block contains the four positive slots, and two unordered pairs partition the four negative slots. There are exactly three pairings. This contributes

$$3\kappa_{40}\kappa_{02}^2 M_4 \overline{M_2}^{\,2}.$$

Its conjugate pattern contributes the complex conjugate. Their sum and A–B subtraction prove (3). For $0 < t < 1$, $\kappa_{40} < 0$ and $\kappa_{20}^2 > 0$. Hence a nonzero D makes the fourth moments different. Bounded random variables with identical laws have identical fourth moments, which proves nonidentity of the laws. □

The real invariant in this theorem is the real diagonal bispectrum of the *clean structure* at frequencies $(2q_0, 2q_0, -4q_0)$. The bispectrum and its relation to triple correlations are established concepts [4]. The invariant is third order in clean Fourier amplitudes; I^4 is a fourth intensity moment, equivalently an eighth field moment. These are different meanings of "order." The theorem is a sufficient condition. If $D = 0$, it does not decide whether higher moments or other frequencies distinguish the pair.

4 An exact nonzero gap for the 63-site pair

Exact evaluation from the phases in Section 2 gives

$$M_{A,2} = \frac{71}{40} + \frac{413}{40}i\sqrt{3}, \qquad M_{B,2} = \frac{131}{8} + \frac{171}{40}i\sqrt{3},$$
$$M_{A,3} = -1 + \frac{1183}{400}i\sqrt{3}, \qquad M_{B,3} = 1 - \frac{1183}{400}i\sqrt{3},$$
$$M_{A,4} = \frac{123901}{48000} - \frac{470783}{48000}i\sqrt{3}, \qquad M_{B,4} = \frac{6145}{384} - \frac{173441}{48000}i\sqrt{3}.$$

The magnitudes agree as required, and $|M_{G,2}|^2 = 129187/400$. Multiplication gives

$$\mathrm{Re}(M_{A,4}\overline{M_{A,2}}^2) = -\frac{455015761}{240000}, \qquad \mathrm{Re}(M_{B,4}\overline{M_{B,2}}^2) = \frac{455015761}{240000}.$$

These real quantities are unchanged by a translation, because translation phases cancel, and by reflection, because reflection conjugates the product. Their unequal values therefore also prove that A and B are not related by a translation or reflection.

Substitution into the general theorem yields the central exact identity:

$$\boxed{\Delta_4(s) = \frac{455015761}{20000}x^4(1-x)^5(x^3+3x^2+6x+6) > 0, \qquad x = e^{-s}, \quad 0 < s < \infty.} \tag{4}$$

The first three raw moments match; consequently the variances and third central moments match, and the fourth central-moment and fourth-cumulant differences equal Δ_4.

At the convenient exact point $s = \log 2$, so $x = 1/2$,

$$v = \frac{62\log 2}{7\pi^2}, \qquad \Delta_4 = \frac{35946245119}{81920000} = 438.79693748779296875.$$

The common variance at this point is $6454387/6400$. The difference in excess kurtosis is therefore

$$\frac{\Delta_4}{(\mathrm{Var}\, I_G)^2} = \frac{35946245119}{83318223091538} \simeq 0.00043143317.$$

The dimensional size of Δ_4 alone is not a measure of classification accuracy. Standardization and probability-distance bounds are needed.

4.1 A bounded witness and its precise optimality class

For $0 \le f \le 1$, $|\mathbb{E}_A f - \mathbb{E}_B f| \le d_{\mathrm{TV}}(P_A, P_B)$. Normalize intensity by $y = I/n^2 \in [0,1]$ and choose

$$f(y) = \frac{1 + T_4(2y-1)}{2} = 64y^4 - 128y^3 + 80y^2 - 16y + 1,$$

where $T_4(z) = 8z^4 - 8z^2 + 1$. Since $T_4(\cos\vartheta) = \cos 4\vartheta$, f lies in $[0,1]$. The first three moment differences cancel, giving

$$d_{\mathrm{TV}}(P_A, P_B) \ge \frac{64\Delta_4(s)}{n^8} > 0. \tag{5}$$

At $s = \log 2$ the right side equals

$$\frac{35946245119}{317639398742426880000} \simeq 1.13166834 \times 10^{-10}.$$

This is a conservative lower bound, not an estimate of actual TV.

The factor 64 is optimal among polynomials of degree at most four constrained to $[0,1]$ throughout the full interval $y \in [0,1]$. Indeed, for any such polynomial g with leading coefficient a_4, let $p(z) = 2g((z+1)/2) - 1$. Then $|p| \le 1$ on $[-1,1]$ and its leading coefficient is $a_4/8$. If $a_4 > 64$, the polynomial $p - (a_4/64)T_4$ has degree at most three, but alternates strictly in sign at the five extrema $z_j = \cos(j\pi/4)$, and so has at least four distinct zeros. This is impossible. Apply the same argument to $-p$ for $a_4 < -64$. Thus $|a_4| \le 64$, and equality is attained by f. This argument does not optimize over all bounded functions or all experiments.

4.2 The unique maximum of this witness gap

Put $C = 455015761/20000$ and view the gap as a polynomial in x. Direct differentiation gives

$$\frac{d\Delta_4}{dx} = 2Cx^3(1-x)^4H(x), \qquad H(x) = 12 - 12x - 21x^2 - 13x^3 - 6x^4.$$

The function H is strictly decreasing on $(0,1)$, with $H(0) = 12$ and $H(1) = -40$. It has a unique zero, which is the unique maximum of the gap and of the bound in (5). Rational bisection encloses the zero in an interval of width at most 2^{-160}, giving

$$x_* \simeq 0.4718312065944692845, \quad s_* \simeq 0.7511339705027228761, \quad v_* \simeq 0.6740797919784131209.$$

The maximum gap is approximately 445.09685324, and the maximum value of the bounded-witness TV lower bound is approximately $1.14791598 \times 10^{-10}$. The exact x_* bracket and rational gap enclosures are certified; the displayed logarithmic conversion to s_*, v_* is descriptive decimal evaluation. No claim about the optimum of full TV, detector performance, or experimental exposure follows from this maximum.

4.3 Endpoint behavior and the importance of the stronger control

The factorization gives

$$\Delta_4(s) = \frac{455015761}{1250}s^5 + O(s^6) \quad (s \downarrow 0), \qquad \Delta_4(s) \sim \frac{1365047283}{10000}e^{-4s} \quad (s \to \infty).$$

At zero motion both intensities are zero. At infinite phase variance the independent wrapped Gaussian phases approach uniform phases, producing the same finite-n random-phasor law for both structures. Equality of those limits is compatible with strict inequality at every finite positive s.

The original graph-dependent controls matched selected marginal properties within each structure but did not impose the same A–B variance profile. Here the entire noise law is identical across the two structures. Moreover, their leading small-noise Gaussian intensity laws are also identical; the new separation is beyond that leading approximation. Therefore the earlier constant lower bounds for the graph-dependent controls must not be transferred to this homogeneous control.

5 Common measurement noise and the fifth-power loss of information

This section derives consequences of the homogeneous-control theorem. These are mathematical measurement models, not experimental validation. The phase variance is $s > 0$, the number of sites is $n = 63$, and

$$X_s = I_A(s), \qquad Y_s = I_B(s), \qquad 0 \le X_s, Y_s \le n^2.$$

Their first three moments agree exactly. Write

$$D_j(s) = \mathbb{E}X_s^j - \mathbb{E}Y_s^j, \qquad C = \frac{455015761}{20000}, \qquad c_5 = 16C = \frac{455015761}{1250}.$$

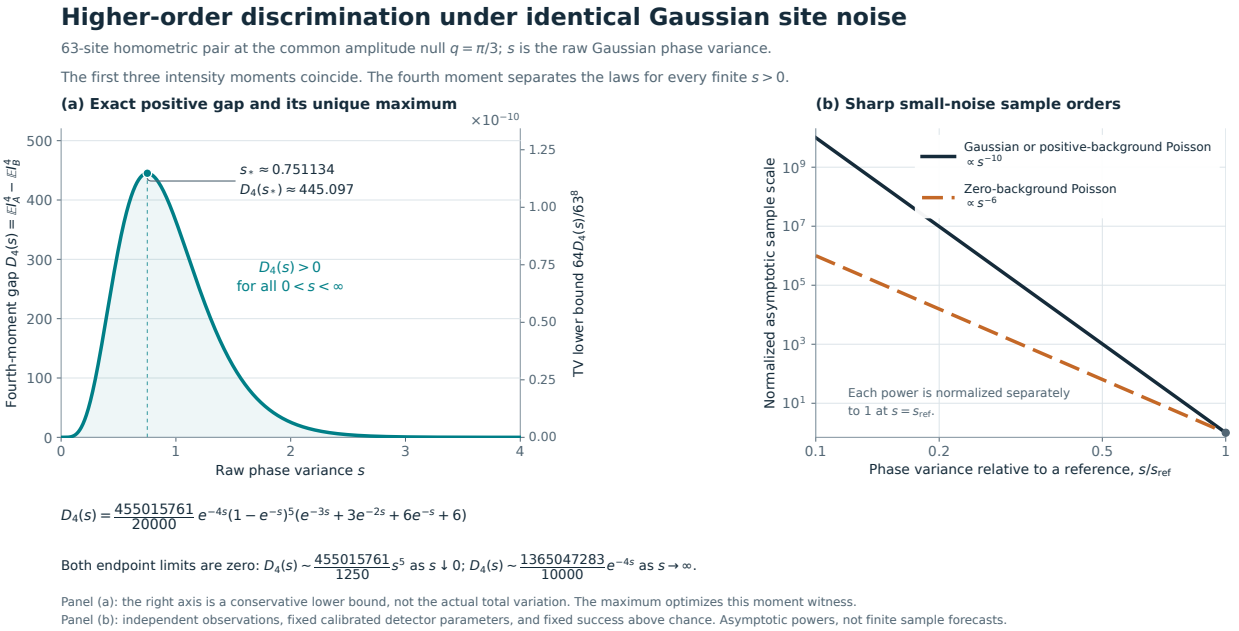


Figure 1: Exact homogeneous-control gap and the asymptotic sampling powers. The TV axis is a conservative lower bound. The normalized sampling curves show powers only; they do not predict finite sample counts.

The exact fourth-moment identity gives

$$D_4(s) = Ce^{-4s}(1 - e^{-s})^5(e^{-3s} + 3e^{-2s} + 6e^{-s} + 6) > 0, \qquad D_4(s) = c_5 s^5 + O(s^6).$$

The separately computed fifth-moment polynomial satisfies $D_5(s) = O(s^6)$. This also follows analytically: the scaled intensities $I_G(s)/s$ tend in every fixed moment to the squared modulus of a complex Gaussian with common covariance and common pseudocovariance magnitude; therefore the s^5 coefficients of the fifth moments agree. Every moment is a finite exponential polynomial, so its Taylor expansion has integer powers. The saved rational calculation verifies all coefficients through order five of D_5 are zero without using this limiting argument.

5.1 A uniform moment bound

Let $Z \sim N(0, I_n)$, $P = I - \mathbf{1}\mathbf{1}^T/n$, and $U = PZ$. Centering changes the amplitude by a common phase and leaves intensity unchanged. At the common clean null, $\sum_i z_i = 0$, so

$$\left|\sum_i z_i e^{i\sqrt{s}U_i}\right| = \left|\sum_i z_i (e^{i\sqrt{s}U_i} - 1)\right| \le \sqrt{ns}\,\|U\|.$$

Since $\|U\|^2$ is chi-square with $n - 1$ degrees of freedom,

$$\mathbb{E}X_s^p + \mathbb{E}Y_s^p \le B_p s^p, \qquad B_p = 2n^p \prod_{j=0}^{p-1} (n - 1 + 2j). \tag{6}$$

The identity for the chi-square moment follows by differentiating its moment generating function $(1 - 2t)^{-(n-1)/2}$ at zero. This estimate is deliberately conservative and holds for every $s > 0$.

5.2 Distinction survives common additive noise

Let E have any fixed distribution with finite fourth absolute moment, independent of the intensity, and use the same noise law for A and B. The binomial expansion proves that the first three moments of $X_s + E$ and $Y_s + E$ agree and

$$\mathbb{E}(X_s + E)^4 - \mathbb{E}(Y_s + E)^4 = D_4(s) > 0.$$

Thus the measured distributions remain distinct at every finite positive s. This is an identifiability statement; it supplies no uniform practical accuracy guarantee as the noise level grows or the signal shrinks.

5.3 Gaussian readout: an exact leading total-variation asymptotic

Take $E \sim N(0, \sigma^2)$ with fixed $\sigma > 0$, and let g_σ be its density. Denote total variation by $d_{\mathrm{TV}}(P, Q) = \frac{1}{2}\|p - q\|_1$ when densities exist. Taylor's theorem with an integral remainder gives, in $L^1(\mathbb{R})$,

$$p_A - p_B = \frac{D_4(s)}{24} g_\sigma^{(4)} - \frac{D_5(s)}{120} g_\sigma^{(5)} + R_s, \qquad \|R_s\|_1 \le \frac{B_6 s^6}{720} \|g_\sigma^{(6)}\|_1.$$

All terms of degree zero through three cancel. Translation invariance of the L^1 norm justifies the remainder bound for all nonnegative input values, including unbounded Taylor segments. Consequently,

$$d_{\mathrm{TV}}(X_s + E, Y_s + E) = \frac{c_5}{48} \|g_\sigma^{(4)}\|_1 s^5 + O(s^6). \tag{7}$$

The remainder may depend on the fixed detector parameter σ. In particular the distance is of order s^5, with a strictly positive coefficient; this statement concerns total variation, not merely one selected statistic.

For explicit constants, let H_k be the probabilists' Hermite polynomial. Then

$$\|g_\sigma^{(k)}\|_1 = \sigma^{-k} \mathbb{E}|H_k(Z)| \le \frac{\sqrt{k!}}{\sigma^k}, \qquad Z \sim N(0, 1).$$

The inequality follows from Cauchy–Schwarz and $\mathbb{E}H_k(Z)^2 = k!$, which follows by repeated Gaussian integration by parts. Put $a = \sqrt{3 - \sqrt{6}}$, $b = \sqrt{3 + \sqrt{6}}$, and $\varphi(t) = e^{-t^2/2}/\sqrt{2\pi}$. Splitting at the four roots of $H_4(t) = t^4 - 6t^2 + 3$ and integrating $(H_3\varphi)' = -H_4\varphi$ yields

$$\|g_\sigma^{(4)}\|_1 = \frac{4\sqrt{6}}{\sigma^4} \big[a\varphi(a) + b\varphi(b)\big] > 0.$$

The saved arithmetic supplies rational A_4, A_5 such that

$$|D_4(s) - c_5 s^5| \le A_4 s^6, \qquad |D_5(s)| \le A_5 s^6.$$

For a polynomial $D_j(s) = \sum_m a_{jm} e^{-ms}$ they are obtained from $A_j = \sum_m |a_{jm}| m^6/6!$ after the exact lower-order cancellations. Thus an explicit finite-s error in (7) is

$$s^6 \left[\frac{A_4 \|g_\sigma^{(4)}\|_1}{48} + \frac{A_5 \|g_\sigma^{(5)}\|_1}{240} + \frac{B_6 \|g_\sigma^{(6)}\|_1}{1440} \right]. \tag{8}$$

A simpler all-s upper bound, using a Taylor expansion only through degree four, is

$$d_{\mathrm{TV}}(X_s + E, Y_s + E) \le \min\left\{1, \frac{D_4(s)}{2\sqrt{24}\,\sigma^4} + \frac{B_5 s^5}{2\sqrt{120}\,\sigma^5}\right\}.$$

5.4 Photon counting: strict distinction and fifth-power scaling

Consider a count N conditional on intensity I with

$$N \mid I \sim \text{Poisson}(b + \eta I), \qquad b \geq 0, \quad \eta > 0,$$

where the background b and gain η are fixed and common to the two hypotheses. The falling factorial $(N)_j = N(N-1)\cdots(N-j+1)$ satisfies $\mathbb{E}[(N)_j \mid I] = (b+\eta I)^j$. Hence the first three factorial moments agree and the fourth differs by $\eta^4 D_4(s) > 0$. The two count distributions are therefore distinct for every finite $s > 0$.

Let p_b be the Poisson mass function at mean b, and let the shift operator S act by $(Sp)(k) = p(k-1)$ with $p(-1) = 0$. Derivatives with respect to intensity satisfy

$$\frac{d^j}{dI^j} p_{b+\eta I} = \eta^j (S - \mathrm{Id})^j p_{b+\eta I}, \qquad \left\| \frac{d^j}{dI^j} p_{b+\eta I} \right\|_1 \leq (2\eta)^j.$$

Taylor expansion in the Banach space ℓ^1 therefore proves

$$d_{\mathrm{TV}}(N_A, N_B) = \frac{c_5 \eta^4}{48} \|(S - \mathrm{Id})^4 p_b\|_1 s^5 + O(s^6). \tag{9}$$

The leading constant is positive: the generating function of $(S - \mathrm{Id})^4 p_b$ is $(z-1)^4 e^{b(z-1)}$, which is not the zero function. For zero background, p_0 is a point mass and the fourth difference has coefficients $(1, -4, 6, -4, 1)$, so

$$d_{\mathrm{TV}}(N_A, N_B) = \frac{c_5 \eta^4}{3} s^5 + O(s^6). \tag{10}$$

An explicit all-s error for this last expression is

$$s^6 \left[\frac{\eta^4 A_4}{3} + \frac{2\eta^5 A_5}{15} + \frac{2\eta^6 B_6}{45} \right].$$

An all-s upper bound requiring no sixth-order quantities is

$$d_{\mathrm{TV}}(N_A, N_B) \leq \min \left\{ 1, \frac{\eta^4 D_4(s)}{3} + \frac{2\eta^5 B_5 s^5}{15} \right\}.$$

5.5 An asymptotically optimal count-parity rule

In the zero-background model, the difference in even-count probabilities is

$$\Pr(N_A \text{ even}) - \Pr(N_B \text{ even}) = \frac{1}{2} \left(\mathbb{E} e^{-2\eta X_s} - \mathbb{E} e^{-2\eta Y_s} \right) = \frac{c_5 \eta^4}{3} s^5 + O(s^6).$$

The same coefficient occurs in (10). With equal prior probabilities and one observation, choosing A on even counts and B on odd counts therefore has advantage over random guessing asymptotic to the optimal advantage: the ratio tends to one as $s \downarrow 0$. This is a one-observation asymptotic statement. Its absolute advantage tends to zero, and the argument does not claim finite-s optimality, optimality for positive background, or optimality for multiple observations.

5.6 Sampling and interpretation

For m independent observations from one unknown configuration with equal priors, a one-observation TV bound $\delta(s)$ implies

$$\Pr(\text{correct}) \leq \frac{1 + \min\{1, m\delta(s)\}}{2}.$$

This elementary inequality gives a necessary $\Omega(s^{-5})$ sample bound. Section 6 improves it to sharp necessary and sufficient orders using Hellinger distance. Unknown gain, background, noise calibration, temporal dependence, and changes in the structural model are outside the stated experiment.

6 Sharp orders for the number of independent measurements

The preceding total-variation calculation concerns one observation. Repeated observations require a separate argument: multiplying the one-observation distance by the number of samples generally loses the correct scaling. Here that gap can be closed for the stated, fully calibrated detector models. All limits in this section hold as $s \downarrow 0$, with detector parameters fixed. Define the squared Hellinger distance and the affinity by

$$h^2(P,Q) = \int (\sqrt{p} - \sqrt{q})^2, \qquad \rho(P,Q) = \int \sqrt{pq} = 1 - \frac{h^2(P,Q)}{2}.$$

Integrals are replaced by sums for count distributions. This convention gives $0 \le h^2 \le 2$.

6.1 Common leading moments at every fixed order

The null condition and the uniform bound in (6) imply

$$I_G(s)/s \;\longrightarrow\; J_G = \left| i \sum_j z_{Gj} Z_j \right|^2$$

almost surely and in every fixed moment. Dominated convergence applies because $I_G(s)/s \le n\|PZ\|^2$, which has every finite moment. The real Gaussian field has covariance eigenvalues

$$\lambda_\pm = \frac{n \pm |M_2|}{2}.$$

They are the same for A and B. Thus J_A and J_B have the same law as $\lambda_+ Z_1^2 + \lambda_- Z_2^2$ with independent standard normals. Write $m_k = \mathbb{E} J_G^k > 0$. Each exact intensity moment is a finite sum of exponentials in s, hence is analytic at zero. Therefore, for every fixed integer $k \ge 1$,

$$\mathbb{E} I_G(s)^k = m_k s^k + O(s^{k+1}), \qquad D_k(s) = O(s^{k+1}). \tag{11}$$

The exact fourth-moment identity specifies the sharper coefficient $D_4(s) = c_5 s^5 + O(s^6)$. Put $R = |M_2|^2 = 129187/400$. The generating function

$$\mathbb{E} e^{tJ_G} = \left[1 - 2nt + (n^2 - R)t^2\right]^{-1/2}$$

gives

$$m_4 = 24n^4 + 72n^2 R + 9R^2 > 0. \tag{12}$$

These facts also justify $D_5 = O(s^6)$ and $D_6 = O(s^7)$ without assuming an uncomputed sixth-order symbolic identity.

6.2 Gaussian readout

For fixed $\sigma > 0$, the squared Hellinger distance between the laws of $I_A + N(0,\sigma^2)$ and $I_B + N(0,\sigma^2)$ satisfies

$$\boxed{h^2 = \frac{c_5^2}{96\sigma^8} s^{10} + o(s^{10}).} \tag{13}$$

Proof. Subtract the common mean $\mu(s) = \mathbb{E}I_G(s) = n(1 - e^{-s}) = O(s)$ from both observations; this translation preserves Hellinger distance. Set $V_G = I_G - \mu$. Then $\mathbb{E}V_G = 0$, $|V_G| \le n^2$, and $\mathbb{E}|V_G|^k = O(s^k)$ for every fixed k. Equality of the first three raw moments gives equality of the first three central moments. The fourth central-moment difference is D_4, and the fifth is $D_5 - 5\mu D_4 = O(s^6)$.

Use the Hilbert norm $\|f\|_g^2 = \int |f(x)|^2/g_\sigma(x)\,dx$. The translated Gaussian kernels obey the exact identity

$$\langle g_\sigma(\cdot - u), g_\sigma(\cdot - v)\rangle_g = e^{uv/\sigma^2}.$$

It follows either by completing the square or direct Gaussian integration. All fixed derivatives of these kernels have finite Hilbert norm, uniformly for $|u| \le n^2$; the derivative norms are an exponential times a polynomial in u. Taylor's theorem in this Hilbert space, through degree five, gives

$$\frac{p_{A,s} - p_{B,s}}{s^5} \longrightarrow \frac{c_5}{24} g_\sigma^{(4)} \quad \text{in } L^2(g_\sigma^{-1}dx).$$

The sixth-order remainder is $O(s^6)$ in that norm by the uniform derivative bound and $\mathbb{E}|V_A|^6 + \mathbb{E}|V_B|^6 = O(s^6)$.

Jensen's inequality supplies the needed denominator control globally:

$$\frac{p_{G,s}(x)}{g_\sigma(x)} = \mathbb{E}\exp\left(\frac{xV_G}{\sigma^2} - \frac{V_G^2}{2\sigma^2}\right) \ge \exp\left(-\frac{\mathbb{E}V_G^2}{2\sigma^2}\right).$$

This lower bound tends to one, and $p_{G,s}(x)/g_\sigma(x) \to 1$ for each fixed x. Hence

$$\frac{h^2}{s^{10}} = \int \frac{[(p_{A,s} - p_{B,s})/s^5]^2}{(\sqrt{p_{A,s}} + \sqrt{p_{B,s}})^2} \longrightarrow \frac{c_5^2}{4 \cdot 24^2} \int \frac{(g_\sigma^{(4)})^2}{g_\sigma}.$$

For completeness, the limit follows from Hilbert-norm convergence of the numerator, which gives L^1 convergence of its weighted square; the extra factor $g_\sigma/(\sqrt{p_A} + \sqrt{p_B})^2$ is uniformly bounded and tends pointwise to $1/4$. Differentiating the kernel identity four times in each variable at zero gives $\int (g_\sigma^{(4)})^2/g_\sigma = 4!/\sigma^8$. Substitution proves (13).

6.3 Photon counting with a fixed positive background

For fixed $b > 0$ and $\eta > 0$ in $N \mid I \sim \text{Poisson}(b + \eta I)$,

$$\boxed{h^2 = \frac{c_5^2 \eta^8}{96 b^4} s^{10} + o(s^{10}).} \tag{14}$$

Proof. Let $p_\lambda(k) = e^{-\lambda}\lambda^k/k!$ and use the Hilbert norm $\|f\|_b^2 = \sum_k |f(k)|^2/p_b(k)$. Direct summation gives

$$\langle p_{b+u}, p_{b+v}\rangle_b = e^{uv/b}.$$

This identity ensures uniform Hilbert bounds for all fixed kernel derivatives on the compact interval $0 \le u \le \eta n^2$. Taylor expansion in intensity through degree five, the exact lower-moment cancellations, and the sixth moment bound give

$$(p_{A,s} - p_{B,s})/s^5 \longrightarrow \frac{c_5 \eta^4}{24}(S - \text{Id})^4 p_b \quad \text{in } \ell^2(p_b^{-1}).$$

For every count k and every admissible intensity,

$$\frac{p_{b+\eta I}(k)}{p_b(k)} = e^{-\eta I}(1 + \eta I/b)^k \ge e^{-\eta n^2}.$$

The mixture ratios tend pointwise to one and have the same positive lower bound. The argument used for Gaussian readout therefore applies to the Hellinger denominator. Differentiating the kernel identity gives

$$\sum_k \frac{[(S-\mathrm{Id})^4 p_b(k)]^2}{p_b(k)} = \frac{4!}{b^4},$$

which proves the stated coefficient. The assumption $b > 0$ is essential; this formula is not uniform as b tends to zero.

6.4 Zero-background counting and the four-count event

At $b = 0$ and fixed $\eta > 0$ the scaling changes:

$$\boxed{h^2 = \frac{\eta^4 c_5^2}{96 m_4} s^6 + O(s^7).} \tag{15}$$

Proof. For each fixed count k,

$$p_{G,s}(k) = \frac{\eta^k}{k!}\mathbb{E}[I_G^k e^{-\eta I_G}] = \frac{\eta^k m_k}{k!} s^k + O(s^{k+1}).$$

For $k = 0$, use $m_0 = 1$. Expanding the remaining exponential and using the exact equality of moments of orders zero through three shows $p_{A,s}(k) - p_{B,s}(k) = O(s^5)$ for $k = 0, 1, 2, 3$. Their Hellinger contributions are respectively $O(s^{10}), O(s^9), O(s^8), O(s^7)$. At $k = 4$, the leading probability and the difference are

$$p_{G,s}(4) = \frac{\eta^4 m_4}{24} s^4 + O(s^5), \qquad p_{A,s}(4) - p_{B,s}(4) = \frac{\eta^4 c_5}{24} s^5 + O(s^6).$$

The second formula uses $D_5 = O(s^6)$ and the sixth-moment remainder. Consequently

$$(\sqrt{p_{A,s}(4)} - \sqrt{p_{B,s}(4)})^2 = \frac{\eta^4 c_5^2}{96 m_4} s^6 + O(s^7).$$

For $k = 5, 6$, equation (11) yields probability differences $O(s^6), O(s^7)$, so their contributions are $O(s^7), O(s^8)$. Finally, the entire remaining tail is bounded without any nonuniform series:

$$\sum_{k \geq 7} (\sqrt{p_{A,s}(k)} - \sqrt{p_{B,s}(k)})^2 \leq \Pr_A(N \geq 7) + \Pr_B(N \geq 7) \leq \frac{\eta^7}{7!}(\mathbb{E} I_A^7 + \mathbb{E} I_B^7) = O(s^7).$$

Here $\mathbf{1}_{\{N \geq 7\}} \leq (N)_7/7!$ supplies the factorial-moment bound. Summing all contributions proves (15).

Keeping only the binary indicator $\mathbf{1}_{\{N=4\}}$ gives the same leading Hellinger coefficient: its complementary event contributes only $O(s^{10})$. Thus a frequency estimate of exactly-four-count observations achieves the optimal sample-order scaling in this limit. This concerns repeated independent observations, not a claim that four-count events dominate one-observation total variation at every noise level.

The earlier parity rule illustrates this distinction. Its even-count probability is $1 - \eta n s + O(s^2)$, and its A–B difference is $(c_5 \eta^4/3) s^5 + O(s^6)$. If only parity is retained, the two Bernoulli laws have

$$h^2_{\text{parity}} = \frac{c_5^2 \eta^7}{36 n} s^9 + O(s^{10}).$$

Parity is asymptotically optimal for one observation in total variation, but discarding count magnitude loses information for repeated measurements.

6.5 From affinity to necessary and sufficient sample orders

Let m observations be independent conditional on the unknown structure, with equal prior probabilities for A and B. The optimal success probability is $(1 + d_{\mathrm{TV}}(P_s^{\otimes m}, Q_s^{\otimes m}))/2$. Direct Cauchy–Schwarz and $\min(p, q) \le \sqrt{pq}$ give

$$1 - \rho(s)^m \le d_{\mathrm{TV}}(P_s^{\otimes m}, Q_s^{\otimes m}) \le \sqrt{1 - \rho(s)^{2m}}.$$

These bounds and $\rho = 1 - h^2/2$ prove that, for any fixed target success $a \in (1/2, 1)$, the minimum sample count has order $1/h^2(s)$. More explicitly, put $r = 2a - 1$. A necessary sample count is

$$m \ge \frac{\log(1 - r^2)}{2 \log \rho(s)},$$

and taking the ceiling of $\log(1 - r)/\log \rho(s)$ is sufficient for the optimal likelihood-ratio test. Therefore the sharp orders are

$$\boxed{\begin{aligned} m_{\mathrm{Gaussian}}(s) &= \Theta(s^{-10}), && \sigma > 0 \text{ fixed}, \\ m_{\mathrm{Poisson}}(s) &= \Theta(s^{-10}), && b > 0,\ \eta > 0 \text{ fixed}, \\ m_{\mathrm{Poisson}}(s) &= \Theta(s^{-6}), && b = 0,\ \eta > 0 \text{ fixed}. \end{aligned}}$$

With only parity retained in the zero-background case, the sharp order becomes $\Theta(s^{-9})$. The powers are proved; a universal finite-noise sample number, an exact optimal prefactor for the minimum sample count, and an experimental acquisition time are not. Calibration, independent snapshots, fixed exposure/gain, and the assumed structural model remain part of the theorem's conditions. Joint limits in b, η, σ, or the observation frequency require a separate analysis.

7 Fresh global crossing proofs for the three earlier controls

7.1 The functions whose roots are counted

Order each structure's sites by mean position and join pairs separated by less than $5/2$, holding this graph fixed during the motion. Exact comparison gives a path on 63 vertices with one additional chord: $(20, 22)$ for A and $(41, 43)$ for B, using zero-based ordinal indices. Let L_G be the graph Laplacian, $C_G = L_G^+$ its pseudoinverse, $\tau_G = \operatorname{tr} C_G$, and $h_{Gi} = 63(C_G)_{ii}/\tau_G$. The three centered covariance coefficients are

$$\begin{aligned} K_{H,G} &= 63 C_G/\tau_G, \\ K_{0,G} &= P \operatorname{diag}(63 h_G/62) P, \\ K_{\star,G} &= P \operatorname{diag}\left[\frac{63}{61}\left(h_G - \frac{1}{62}\mathbf{1}\right)\right] P. \end{aligned}$$

Their traces are 63, and the actual covariance is vK_G. $K_\star$ has the same diagonal as K_H within each configuration. The whole A–B covariance equality used in the new theorem is a separate, stronger control.

For any such centered Gaussian covariance,

$$\mathbb{E} I_G(v) = 63 + 2 \sum_{i<j} \operatorname{Re}(z_{Gi} \bar{z}_{Gj}) \exp\left[-\frac{q_0^2 v}{2}(K_{G,ii} + K_{G,jj} - 2K_{G,ij})\right].$$

Subtract B from A and use the common coordinate $y = 7\pi^2 v/62$. Exact grouping gives a 2005-term exponential polynomial $D_0(y)$ for the original control and a 203-term polynomial $D_H(y)$ for the harmonic control. All rates are rational; all coefficients are real algebraic numbers in the specified radical field. No

numerical threshold is used to decide whether a coefficient vanishes. The marginal-preserving identity is exact:

$$\Delta_\star(v) = e^{7\pi^2 v/(62\cdot 61)} D_0(7\pi^2 v/61). \tag{16}$$

Its roots are exactly 61/62 times the original roots in v.

Control	Positive roots	Locations in centered MSD v (rounded)
Original	Exactly 2	9.503072132963126, 1956.788193416293
Marginal-preserving	Exactly 2	9.349796775979850, 1925.227093522482
Harmonic	Exactly 1	111.313650423149122

Every positive root is simple. The signs of A–B mean contrast are $+, -, +$ for the first two controls and $+, -$ for the harmonic control. The origin is a double zero for the first two and a simple zero for the harmonic one. The JSON certificates contain exact rational brackets much narrower than the displayed decimal precision.

7.2 How the original and marginal-preserving counts are proved

Let $r_0 = 121483/242972$ be the slowest rate of D_0 and set $F(y) = e^{r_0 y} D_0(y)$. This positive multiplier preserves signs, zeros, and multiplicities. Write

$$F(y) = \sum_i c_i e^{-\lambda_i y}, \quad \lambda_0 = 0, \quad c_0 = \frac{\sqrt{573} - 1}{60} > 0, \quad c_1 = -\frac{146}{75}, \quad \lambda_1 = \frac{189}{242972}.$$

The certificate establishes the following exhaustive partition:

Region	Certified statement
$0 < y \le 1$	$F(y)/y^2 > 17/1000$
$[1, 10]$	$F > 0$ on three closed interval cells
$[10, 11]$	$F' < 0$, $F(10) > 0$, $F(11) < 0$
$[11, 2048]$	$F < 0$ on nine closed interval cells
$[2048, \infty)$	$F' > e^{-\lambda_1 y}/1000$, $\lim F = c_0 > 0$

Exact coefficientwise cancellations give $F(0) = F'(0) = 0$. The divided Taylor bound proves the double zero and covers arbitrarily small positive y. Strict decrease gives exactly one simple root in $(10, 11)$. The negative finite cover, strict tail increase, and positive limiting value give exactly one additional simple root. There is no uncovered region. Equation (16) transfers the complete result to the marginal-preserving model.

7.3 How the harmonic count is proved

The harmonic pair rate in coordinate y is

$$r_{G,ij} = \frac{31}{\tau_G}(C_{G,ii} + C_{G,jj} - 2C_{G,ij}).$$

The 3906 signed pair contributions reduce to 203 nonzero groups. Its slowest coefficient is $-3/5$ at rate 1953/60785, followed by $+3/5$ at rate 1953/60743. Positive rescaling by the slowest exponential again preserves all zeros and signs. The new certificate covers $(0, 1/32]$ by a divided Taylor bound with positive sign, $[1/32, 124]$ by 13 positive cells, $[124, 125]$ by strict negative derivative and opposite endpoint signs, $[125, 512]$ by four negative cells, and the whole tail $[512, \infty)$ by a strict negative bound. The origin is a

simple zero with positive derivative. These facts prove one and only one positive root, and prove its simplicity. The stored bracket is

$$111.313650423149121934480095207692375398084 3 < v_H,$$
$$v_H < 111.313650423149121934480095207692375398759 6.$$

The standalone harmonic proof records the exact origin and tail inequalities, group reconstruction, complete cells, and all source/input bindings.

7.4 Arithmetic and reproducibility of the global certificates

All decision-making interval endpoints are integers divided by 2^{256}. Radicals use integer square roots. Exponentials use alternating Taylor bounds after range reduction, followed by outward squaring. Exponentials below the working unit are enclosed by a nonzero upper bound, not discarded. An exact inequality $e^{7/10} > 2$ justifies the tiny-exponential shortcut. Machin's identity with alternating arctangent bounds encloses π for the coordinate conversion.

For a cell $[a, b]$ with midpoint m and radius h, let

$$C_k(m) = \sum_i c_i e^{-\lambda_i m} \frac{(-\lambda_i)^k}{k!}, \qquad B_j(a) = \sum_i |c_i| e^{-\lambda_i a} \frac{\lambda_i^j}{j!}.$$

Then Taylor's theorem bounds the remainder for F by $B_{p+1}(a)h^{p+1}$ and the remainder for its degree-p derivative expansion by $(p+2)B_{p+2}(a)h^{p+1}$. These bounds apply to the whole cell. Exact rational adjacency checks establish complete coverage. Infinite tails are handled analytically by dominance of the leading signed terms; no finite plot is used as a tail proof.

The source, input, and output hashes allow the exact bytes to be identified. Hashes establish file identity, not theorem correctness. The original/star certificate was separately replayed, including every accepted cell and both root bisections, and its grouped input was independently reconstructed from all pair contributions. The harmonic certificate has its own separate review record. Full proof texts and replay scripts accompany this report.

8 Relationship to earlier work and the remaining novelty question

The broad scientific ideas have substantial precedent. Random displacement transformations of pair correlations and mean diffraction were studied by Gabrielli and by Klatt, Kim, and Torquato [1, 2]. Ravy showed that intensity statistics can reveal distinctions hidden in average diffraction for diffuse-homometric systems [3]. Fourier products at frequencies summing to zero, including bispectral invariants, are central to established higher-correlation reconstruction theory [4]. These sources prevent a claim that fluctuations, homometry, or the invariant itself have just been discovered here.

Finite nonuniform-phase phasor statistics also have direct predecessors. Barakat studied intensity and phase statistics for finite nonuniform-phase models and photoelectron statistics for an arbitrary number of weak scatterers [5, 6, 7]. The publisher abstracts were available, but the full papers were not retrieved in this review; their unread detailed formulas are not claimed to be different from ours. More recent work includes broad large-system phasor laws [8]. Shimizu's wrapped-normal model permits deterministic phase offsets and includes an exact finite characteristic-function construction in its appendix alongside Gaussian many-component limits [9]. Thus exact finite Gaussian-phase algebra itself is established methodology.

The defensible candidate contribution is narrower and explicit: the null selection theorem, its exact noncongruent finite example under an identical full noise law, the coefficient that is nonzero for every finite positive noise level, and the resulting detector and sampling consequences for this particular

comparison. The detection arguments use standard Taylor, Hilbert-space, and hypothesis-testing tools. Applying them here produces proved model statements; it does not by itself establish method priority.

The literature note records the examined assumptions, read depth, and remaining access gaps. No matching statement was established in the sources examined, but a targeted search cannot prove universal absence. Before a world-first claim, the exact audited theorem should be compared with the full finite-phasor and photon-counting literature, including the identified older papers, and assessed by a domain expert. No external messages or submission have been made.

9 What this record closes and what would require new evidence

This record closes the positive-axis crossing counts for the three specified controls, proves a stronger common-motion distributional distinction, and gives sharp small-noise sample orders for three calibrated detector regimes. The source calculations use exact arithmetic for the algebraic statements. The crossing certificates cover the whole positive axis. The detector and sampling results are analytic proofs with explicit conditions, not fitted simulation exponents.

The remaining scientific questions require a change in evidence or scope. Physical usefulness needs an actual scattering implementation, known displacement statistics, detector gain and background, a frequency and exposure choice, and independent or explicitly modeled temporal samples. Unknown calibration parameters could imitate or obscure a small higher-moment effect. Those are experimental and statistical modeling questions; an additional root digit cannot resolve them.

The best finite-noise detector setting, actual unobserved intensity TV, joint low-background limits, and robustness to structural perturbations are not characterized globally here. The exact maximum derived above is only for the bounded fourth-degree witness. No exact count of standardized contrast maxima is claimed; the earlier existence result of at least three maxima remains distinct from a uniqueness or optimal-design theorem.

The new global certificates replace the earlier missing-proof status for the three mean-crossing claims. They are fresh proofs for the reconstructed model; they do not restore historical certificate bytes or the execution history of a prior session. The older immutable report and download have not been altered. Its broader limitations remain in force except where this supplement explicitly supplies new proofs.

A Exact arithmetic values and replay guide

The detector Taylor bounds use

$$A_4 = \frac{66213438524959}{144000}, \qquad A_5 = \frac{347797653430767447163}{192000000},$$

$$B_5 = 2474316747472711680, \qquad B_6 = 11223500766536220180480.$$

The exact coefficients of s^6 in D_4, D_5 are respectively $-54146875559/20000$ and $28665992943/50$. The common leading fourth moment and representative Hellinger constants are

$$m_4 = \frac{75408578015121}{160000}, \qquad \frac{c_5^2}{96} = \frac{207039342758409121}{150000000},$$

$$\frac{c_5^2}{96m_4} = \frac{414078685516818242}{141391083778351875}, \qquad \frac{c_5^2}{36n} = \frac{207039342758409121}{3543750000}.$$

The principal files are organized as follows. The manifest lists exact SHA-256 digests and byte sizes. Python verification scripts require only the standard library, except for the optional plotting script.

Homogeneous theorem. `homogeneous_control/homogeneous_control_proof.txt` gives the full plain-text proof. Running `homogeneous_control/homogeneous_control.py` rebuilds the phases, all moment polynomials through order five, and an independent direct-site convolution check at $t = 1/2$. It also records selected higher-moment values for the bounded statistic.

Detectors and optimization. `detector_proofs.tex` and `sample_complexity_proofs.tex` are the proof sections included verbatim here. Running `measurement_and_optimization.py` checks exact Taylor cancellations and constants, certifies the unique bounded-witness optimum, and writes `measurement_and_optimization_results.json`.

Original and marginal-preserving roots. Run `root_certificate/certify_two_roots.py` beside its unchanged `input_premises.json`. The proof is `two_root_proof.txt`, and `two_root_certificate.json` records the complete cover and brackets.

Harmonic root. `harmonic_certificate/build_harmonic_groups.py` rebuilds the harmonic groups from the copied exact model. Run `harmonic_certificate/certify_harmonic_root.py` to reproduce the global certificate. Its directory includes the proof and input bindings.

Internal review and prior work. `independent_completion_review.txt`, the two root-review notes, and the literature comparison record the review scope and remaining limits.

Reproduction verifies these supplied model definitions and proof artifacts. Agreement between programs is not a substitute for checking the assumptions against a physical system. A self-contained saved record should retain the PDF, proof sources, scripts, their stated input JSON files, review notes, and the manifest together.

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